# SimpliSpeak: User-Controlled Text Simplification for Audio-First Accessible Reading

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## Abstract

Screen readers make digital text available through speech but they do not necessarily make complex language easier to understand. This can create difficulties when blind and low-vision users encounter dense medical, legal, financial or administrative documents. SimpliSpeak is a proposed audio-first reading interface that allows listeners to choose the linguistic complexity of text while listening. Its central design principle is that the listener, rather than the system controls how simple the content should become. The Phase 1 project produced an eleven-screen low-fidelity prototype and a set of manually prepared examples. The proposed interaction includes four text levels (Original, Light, Simple, and Very Simple) short-part playback, on-demand word explanations, navigation and repetition and access to the exact original text. Text can be introduced through clipboard input, scanning or a text field. This report describes the design rationale, conceptual and proposed system architecture, prototype scope, implementation compromises, evaluation plan, and ethical and safety considerations. The current prototype is not an operational AI application: automatic simplification, optical character recognition and real audio playback have not been implemented and no user evaluation has yet been conducted. Accordingly, this report presents the evaluation protocol and outcomes to be measured rather than claiming empirical results.

## CCS Concepts

• **Human-centered computing → Accessibility**.

## Keywords

accessible computing, blind and low-vision users, text simplification, screen readers, audio-first interfaces, human–computer interaction

## 1 Introduction

Digital information is increasingly distributed through websites, applications and electronic documents. Screen readers provide access to this information by converting displayed text into speech. Access to the words does not guarantee that the information is easy to understand. A screen reader generally speaks the text as it is written; when a document contains long sentences, unfamiliar terminology or dense instructions, the same linguistic difficulty remains in the audio presentation.

This challenge is especially relevant to medical leaflets, bank communications, legal notices and other documents where users may need to understand and act on specific information. Unlike visual reading, listening is sequential: spoken words disappear as they are heard and listeners may find it difficult to skim ahead or quickly revisit an earlier phrase. Research on screen-reader browsing has also described non-linear navigation strategies among blind users, rather than assuming that all content is consumed from beginning to end [3]. Research on listening speeds among blind users further indicates that listening practices can differ from typical speech rates [1]. These considerations suggest that an accessible reading interface should support both efficient navigation and comprehension.

SimpliSpeak explores an audio-first approach to this problem. It is designed to simplify text in short portions while allowing the listener to choose the level of simplification. The listener can request a simpler version, ask for an explanation of a difficult word repeat or navigate between portions, and return to the exact original wording. The guiding principle is: *the listener, not the AI, decides how simple the text should be.*

It demonstrates the proposed screens and interaction flow, rather than a fully implemented application. The report therefore distinguishes between:

(1) functionality represented in the prototype,
(2) functionality manually demonstrated using prepared examples, and
(3) functionality proposed for a later implementation.

## 2 Conceptual Design

### 2.1 Design Problem and Target Users

The design addresses the gap between making text audible and making its meaning accessible. The intended users are blind and low-vision people who use audio interfaces to access written information. The design also considers differences in experience with assistive technology. An experienced screen-reader user may want to navigate quickly and independently, while a person who is newly adapting to audio tools may benefit from slower speech, repetition and straightforward ways to request help.

Two illustrative personas informed the initial design. Asha represents an experienced screen-reader user who can operate her phone independently but may still need help understanding dense bank letters or medicine leaflets. Ravi represents a person who has recently lost sight and is learning audio tools while managing health-related paperwork. These personas were developed from background reading and team discussion; they are design aids, not findings from interviews.

### 2.2 Design Goals

Five user needs informed the design:

(1) **Understand without repeatedly restarting:** present content in short portions and make repetition and backward navigation available.
(2) **Retain control over simplification:** provide several levels of linguistic complexity and let the listener choose or change the level.
(3) **Trust transformed content:** keep the exact source text accessible and protect critical information from alteration.
(4) **Resolve unfamiliar vocabulary:** explain difficult words only when the listener requests help.
(5) **Interact without relying on sight:** support voice, gesture, keyboard, and concise audio cues.

These goals are consistent with broader work on cognitive load and plain-language accessibility. Sweller's work on cognitive load motivates attention to the amount of information that must be held and processed at once [8]. WCAG 2.2 includes a reading-level criterion for certain content, while W3C's cognitive and learning disabilities guidance emphasises making content more understandable [5, 9]. Research on text simplification provides a broader foundation for transforming complex language into more accessible forms [7].

### 2.3 Core Interaction Concept

The primary interaction is a repeated listen–adjust loop. A user brings text into the application, selects a preferred level, listens to one short portion at a time and requests an adjustment when needed. The user can ask for simpler wording, more detail, an explanation or repetition. The original text remains available throughout. The intended interaction is not to replace the source document, but to provide an additional, listener-controlled way to understand it.

## 3 System Design and Architecture

### 3.1 Proposed Architecture

The Phase 1 deliverable is a low-fidelity interface prototype, not a functioning software system. The architecture below describes the proposed structure for a future implementation and should not be interpreted as components already integrated or tested.

The proposed system can be organised into five logical layers:

(1) **Input layer:** receives text through clipboard paste, a text-entry field or camera-based scanning. Document and share-sheet integration are future extensions.
(2) **Text preparation layer:** extracts text from the selected source, segments it into manageable portions, and retains the original text as an immutable reference.
(3) **Simplification and explanation layer:** produces the requested level (Original, Light, Simple, or Very Simple) and generates on-demand explanations for selected words or phrases.
(4) **Audio and interaction layer:** presents the current portion through speech output and receives commands for play/pause, next/back, repeat, simplification level, and word explanation.
(5) **Safety and state-management layer:** tracks the current reading position, stores the source text, checks protected information, and provides a fallback to the original text when a transformation is uncertain or fails.

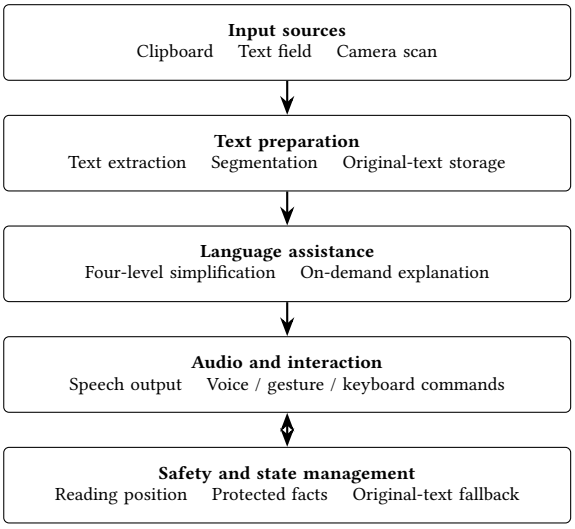



**Figure 1: Conceptual architecture proposed for a future SimpliSpeak implementation. The Phase 1 prototype represents the interaction flow but does not implement these processing services.**

### 3.2 Input and Content Processing

The prototype presents three immediate ways to provide text: paste from the clipboard, scan printed material or enter text in a text box. Clipboard input addresses the situation in which a user has already copied text from another application. Scanning addresses the situation in which the user has a printed document. The text box provides a direct entry route. Opening files such as PDF and Word documents and receiving content through email or a share sheet were deferred because of the Phase 1 timeline.

In a future implementation, extracted text should be segmented into short, navigable portions. The system must preserve a mapping between each simplified portion and its corresponding source text. This mapping is necessary to support returning to the original wording and changing the simplification level without losing the current reading position.

### 3.3 Simplification Levels and Interaction State

SimpliSpeak proposes four levels:

- **Original:** the source wording, unchanged.
- **Light:** modest changes to difficult wording while retaining much of the original sentence structure.
- **Simple:** more direct wording and shorter sentence structures.
- **Very Simple:** highly accessible wording, potentially separating a long sentence into several short statements.

Changing levels is intended to preserve the current sentence or content position. The listener should not have to restart the document or manually locate the same passage. The interaction state therefore needs to track the document, current segment, selected level, playback state and any requested explanation. These

are requirements for the proposed application; they are not implemented as a working state-management system in the current Figma prototype.

### 3.4 Audio and Non-Visual Controls

The proposed listening screen supports multiple ways to issue commands. Play/pause is associated with a double tap, a spoken command or the Space key. Next and back are associated with horizontal swipes, spoken commands or arrow keys. Simpler and more detail are associated with vertical swipes, spoken commands or arrow keys. Explain is associated with a long press, a spoken command or the E key. Repeat is associated with a spoken command or the R key. The prototype also proposes speech-speed adjustment up to 1.8× and short sound cues to communicate system state without adding unnecessary spoken instructions. Earcons have been studied as a way of conveying information through brief auditory signals [2]. These mappings require usability and accessibility testing before they can be treated as validated interaction conventions.

### 3.5 Safety and Source Fidelity

The design identifies doses, dates, amounts, names and warnings as protected information that should not be altered during simplification. The exact original text must remain available on demand. If a future AI system cannot confidently simplify a passage without changing its meaning, it should communicate that limitation and offer the source wording instead. This is particularly important for health and financial content, where a plausible but inaccurate rewrite could lead to harmful decisions. The current prototype communicates this requirement through design examples; it does not yet contain an automated verification mechanism.

## 4 Prototype Description and Functionalities

### 4.1 Prototype Overview

Phase 1 produced an eleven-screen low-fidelity prototype in Figma. The prototype is intended to communicate the main screen structure, navigation, and interaction logic. It includes manually prepared simplified wording for three sample texts and demonstrates the intended use of clipboard input, scanning, a text box, and voice, gesture, and keyboard controls. It does not yet provide automatic AI simplification, real audio playback, or complete document processing.

### 4.2 Main Functionalities

The prototype is organised around four core features:

(1) **Short-part reading:** text is presented in short portions, with options to repeat, move forward or return to an earlier portion.
(2) **Listener-selected simplification:** users can choose Original, Light, Simple, or Very Simple. The design intends that a level change retains the current sentence.
(3) **On-demand word help:** the user requests an explanation of a difficult word rather than receiving unsolicited explanations during playback.
(4) **Original text access:** the exact source wording remains one command away, allowing users to verify a simplified version.

The central workflow is: bring text into the application, choose a level, listen to one short portion at a time, adjust the wording or request help, and continue until the content is clear. The loop between listening and adjustment is central to the design.

### 4.3 Illustrative Example

The prototype uses a medicine-leaflet warning to demonstrate the four levels. The original warning asks the reader to discontinue use and seek immediate medical attention if they experience difficulty breathing, facial swelling or severe skin reactions. The Light, Simple, and Very Simple examples use shorter and more direct wording while attempting to retain the same key actions and symptoms. The example illustrates the design requirement that simplification should not remove the instruction to stop the medicine or the warning signs that require medical attention. It is a manually prepared example, not evidence that an AI model has reliably preserved the meaning.

### 4.4 Prototype Access

The Figma prototype is available at:
https://www.figma.com/design/U6Hc0EEJhZeWv5MggR3LUN/SimpliSpeak-%E2%80%93-Low-fi-prototype--HCI-Phase-1-?node-id=0-1&p=f&t=0sQc12bqsXLLvQtW-0

## 5 Compromises in Prototyping

The project deliberately used a low-fidelity prototype to communicate the interaction concept within the available time. This allowed the team to focus on the organisation of the reading flow, the placement and naming of controls, and the relationship between simplification levels and listening position before investing in a full technical implementation.

Several compromises were made:

- **Manual rather than automatic simplification:** simplified versions were prepared manually for three examples. This made it possible to demonstrate the intended difference between levels without integrating an AI model. It does not test the quality, reliability or scalability of automatic simplification.
- **Simulated rather than real speech:** the prototype communicates the intended audio interaction but does not implement a working text-to-speech pipeline. Consequently, speech timing, voice quality, latency and listening effort cannot yet be evaluated.
- **Represented rather than implemented scanning:** the scan route is part of the proposed input flow, but OCR and camera processing are not implemented in the current prototype.
- **Limited document integration:** clipboard, scan, and text-entry routes are represented. PDF and Word opening, email integration, and share-sheet integration were deferred because of the project timeline.
- **Unvalidated interaction mappings:** voice, gesture, and keyboard commands are proposed to support non-visual use.

Their discoverability, consistency and compatibility with existing screen-reader gestures require testing with intended users.
- **Illustrative personas:** the initial personas were based on reading and team discussion, not direct interviews. They should therefore guide initial design decisions but should not be treated as representative user evidence.

These compromises are appropriate for an early design prototype, but they define what can and cannot be concluded from Phase 1. The current work demonstrates a proposed interaction model; it does not establish that the system improves comprehension, reduces workload, or is usable by blind and low-vision users.

## 6 User Evaluation and Results

### 6.1 Evaluation Status

No user evaluation has been completed for the current Phase 1 prototype. The following is the planned evaluation protocol. Participant numbers and recruitment arrangements remain to be finalised. No usability scores, comprehension outcomes, or participant preferences are reported in this paper.

### 6.2 Evaluation Objectives

The proposed evaluation will examine six questions:

(1) Can participants complete the core tasks without looking at the interface?
(2) Does simplified wording support better comprehension than the original wording?
(3) Does the interaction impose acceptable perceived workload?
(4) Do participants find the interface usable and trustworthy?
(5) Which simplification levels do participants choose for different documents?
(6) Do participants prefer clipboard input or scanning for the same task?

### 6.3 Participants and Procedure

The intended participant group is blind and low-vision people who use screen readers or other audio-accessibility tools. The final sample size, eligibility criteria, recruitment process, and accessibility arrangements must be determined before the study begins. Participants should be able to use their preferred assistive technology during testing, and the study should not assume that all participants have the same level of experience with screen readers.

A proposed session would introduce the study and obtain informed consent, followed by a short orientation to the prototype. Participants would then complete tasks involving bringing text into the system, choosing a reading level, listening to a portion, requesting simplification or an explanation, repeating or navigating, and returning to the original text. Where feasible, participants would also complete comparable tasks using clipboard input and scanning. The evaluator would record task completion, errors, time and verbal feedback. Think-aloud instructions should be adapted so that they do not interfere with listening or create unnecessary cognitive burden.

**Table 1: Planned user-evaluation measures**

| Evaluation question | Measure |
|---|---|
| Task completion | Completion rate, time on task, navigation errors, and requests for assistance. |
| Comprehension | Accuracy on questions about content presented at each simplification level. |
| Perceived workload | NASA Task Load Index (NASA-TLX) after each relevant task or text condition. |
| Usability and trust | System Usability Scale (SUS), followed by a short interview about confidence, control, and perceived changes in meaning. |
| Level selection | Frequency and sequence of level changes across documents or tasks. |
| Input preference | Task time, errors, preference, and comments for clipboard and scan routes. |

### 6.4 Measures

The proposed measures are summarised in Table 1. The System Usability Scale (SUS) provides a standardised measure of perceived usability [4], while NASA-TLX can be used to assess perceived workload [6]. Comprehension questions should be designed to test important information in the source material, not merely recall of wording. For safety-related examples, questions should include critical facts such as instructions, dates, amounts, and warning signs.

### 6.5 Results and Reporting Plan

Because testing has not yet been conducted, there are no empirical results to report. After data collection, results should be presented descriptively and where the sample and study design justify it, with appropriate statistical comparisons. The report should include participant characteristics in non-identifying form, task-completion rates, errors, comprehension accuracy, workload and usability scores, level-selection patterns and qualitative themes from interviews. Any exclusions, missing data, or deviations from the planned procedure should be documented. Results should not be interpreted as evidence of improved independence or safety unless the study directly measures those outcomes.

## 7 Ethics and Risk Assessment

SimpliSpeak is intended to support access to written information, including potentially sensitive medical, legal and financial content. This creates ethical responsibilities beyond ordinary interface usability.

### 7.1 Meaning Distortion and High-Stakes Errors

The most important risk is that simplification may change or omit information. In medical content, an altered dose, symptom, contraindication, or instruction could cause harm. In financial or legal content, changing an amount, deadline, obligation, or condition could lead to a poor decision. The design therefore treats doses, dates, amounts, names, and warnings as protected facts. A future implementation should preserve the original text, provide an easy route back to it, and avoid presenting uncertain transformations as

reliable. Human review may be necessary for high-stakes use. SimpliSpeak should not be positioned as a replacement for professional medical, legal, or financial advice.

## 7.2 Privacy and Data Protection

Documents may contain names, addresses, account information, health details, or other personal data. A future implementation should minimise collection, explain whether text is processed locally or sent to an external service, obtain appropriate consent, secure data in transit and at rest, and define retention and deletion practices. The prototype does not currently process or store real user documents, but these requirements must be addressed before implementation with real data.

## 7.3 Accessibility and Inclusion

An application intended for blind and low-vision users must not depend on visual feedback for essential actions. Controls should have accessible names, predictable keyboard operation, compatibility with commonly used screen readers, and clear audio feedback. Gesture mappings must be checked for conflicts with operating-system and screen-reader gestures. Evaluation should include participants with different levels of vision, assistive-technology experience and interaction preferences. The team should avoid assuming that a single audio interaction style works for everyone.

## 7.4 Consent, Participant Burden, and Representation

Future user studies should use accessible participant information and consent materials. Participation should be voluntary and participants should be able to pause or withdraw without penalty. Tasks involving medical or financial documents should use fictional or carefully sanitised material unless real documents are essential and appropriate safeguards are in place. Participants should not be asked to disclose diagnoses or personal information that is not necessary for the research. The personas used in Phase 1 should be treated as provisional until checked through direct engagement with intended users.

## 7.5 Risk Mitigation Summary

Table 2 summarises the principal risks and the proposed mitigations.

**Table 2: Principal risks and proposed mitigations**

| Risk | Proposed mitigation |
|---|---|
| Meaning changed or critical fact omitted | Preserve source text; protect critical entities and instructions; compare output with source; provide original-text fallback; flag uncertainty. |
| Incorrect OCR or text extraction | Allow users to review or replay extracted text; indicate extraction failures; avoid silently processing uncertain content. |
| Sensitive document exposure | Minimise data collection; disclose processing and retention; use secure transmission and storage; provide deletion controls. |
| Inaccessible controls or feedback | Use screen-reader labels, keyboard support, non-visual status feedback and testing with blind and low-vision users. |
| Cognitive overload from excessive prompts | Keep assistance on demand; make prompts concise; allow users to control speech speed and navigation. |
| Over-reliance on simplified output | Keep original wording available; communicate limitations; do not present the tool as professional advice. |

# 8 Conclusion

SimpliSpeak explores an audio-first approach to making complex written information easier to understand for blind and low-vision users. Its central design contribution is a listener-controlled interaction in which users can select among four levels of text complexity, request explanations when needed, navigate through short portions and return to the original wording without losing their place.

The current prototype is an early design artefact rather than a functioning AI reading application. Automatic text simplification, OCR, document processing, and real audio playback remain future implementation tasks. Likewise, the proposed benefits for comprehension, workload, usability, and trust have not yet been established through user evaluation. The next stage should implement the core audio and simplification pipeline, establish safeguards for preserving critical information, and evaluate the interaction with blind and low-vision users. SimpliSpeak's success should ultimately be judged not only by whether it can simplify text, but by whether it helps listeners understand information while retaining control and confidence in what they hear.

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